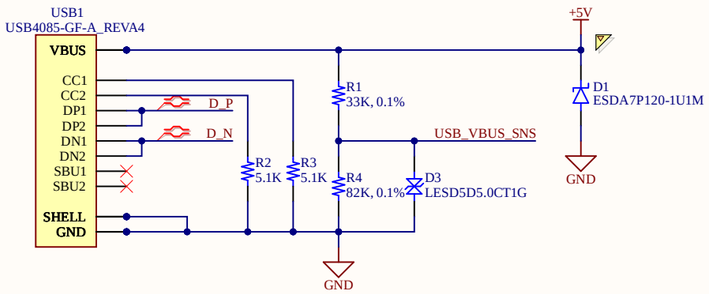
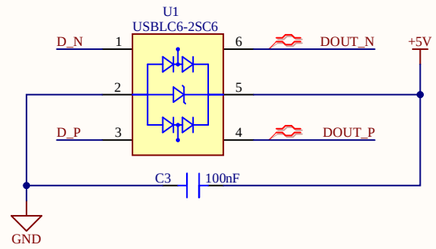
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

POWER SCHEME

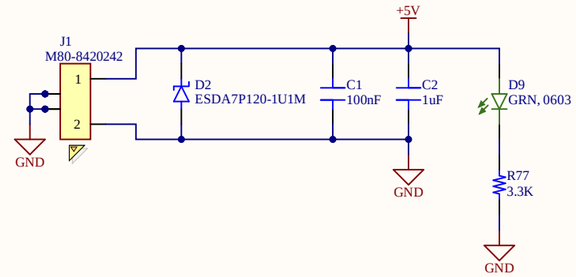
USB-C



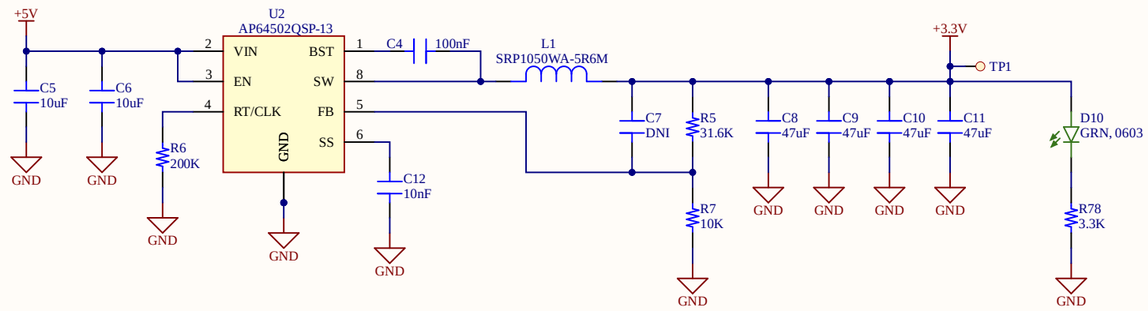
USB ESD



+5V FROM PDB



SWITCHING REGULATOR

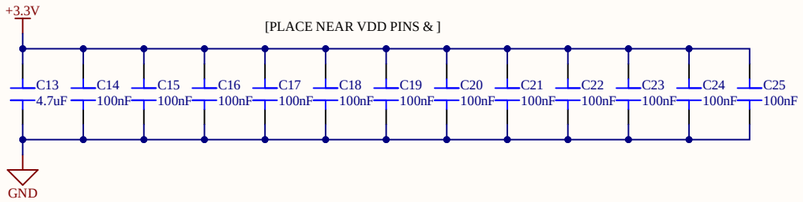


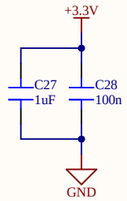
SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 2 OF 7	

UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

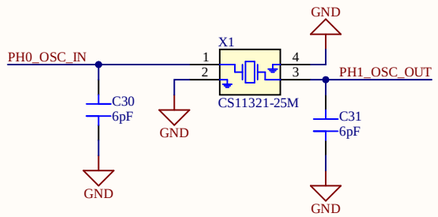
STM32H743ZIT6

DECOUPLING STM32H7

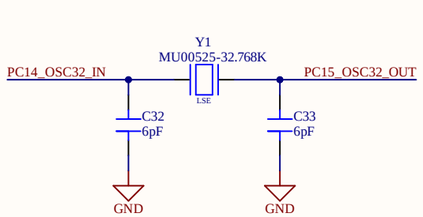




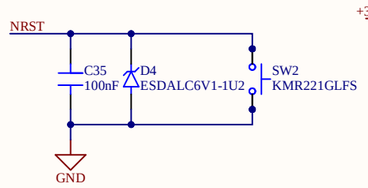
HSE

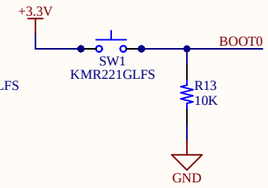


LSE

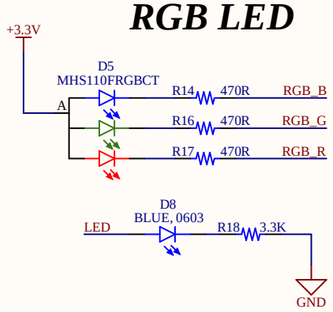


NRST & BOOT

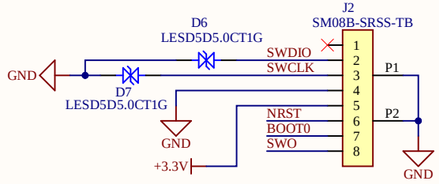


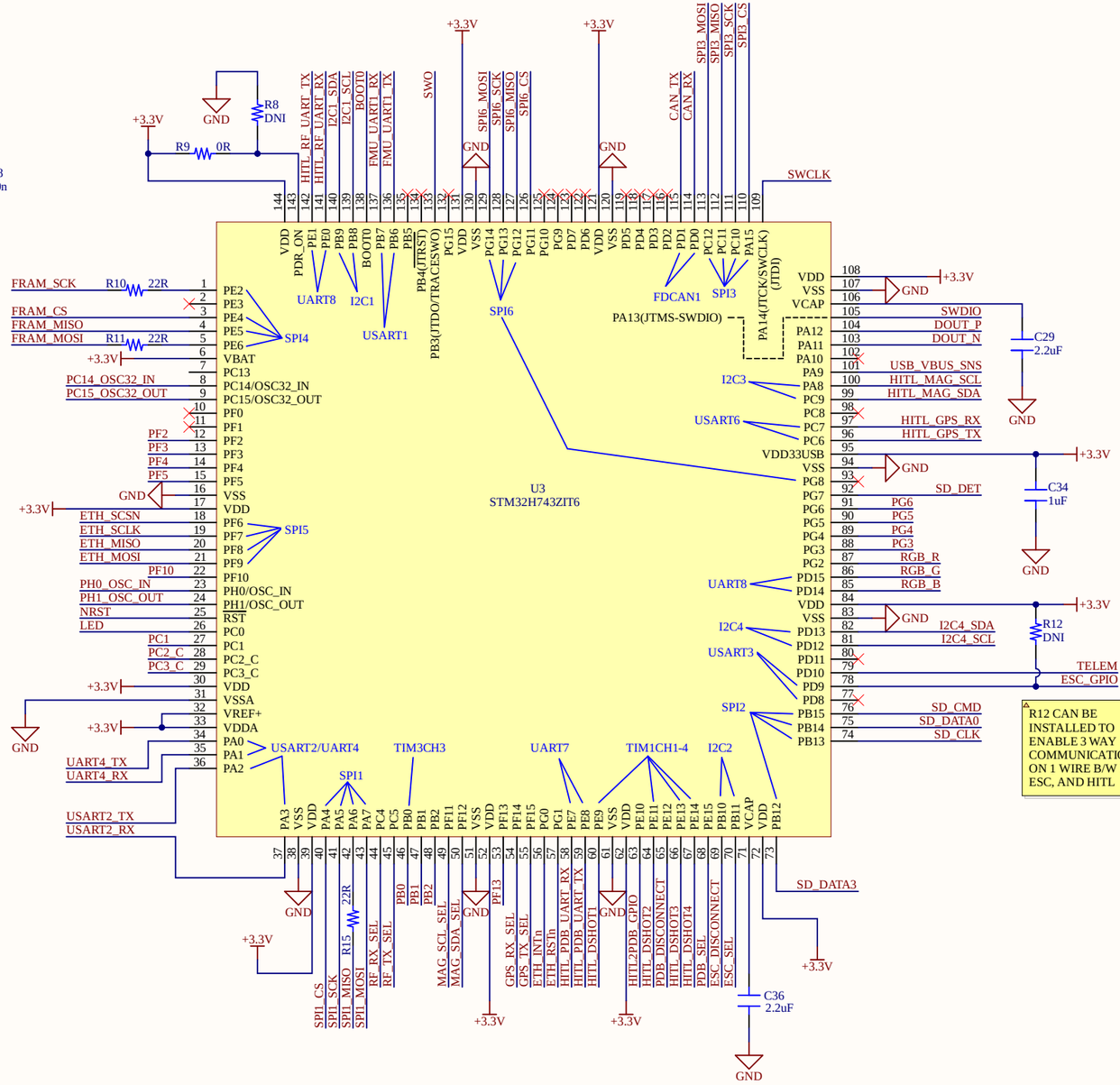


RGB LED



PRGM CONN





	SIZE B	DWG NO	REV -
SCALE:	VIEWDRAW	SHEET 3 OF 7	

LOGIC LOW == SYS NORMAL FUNCTION

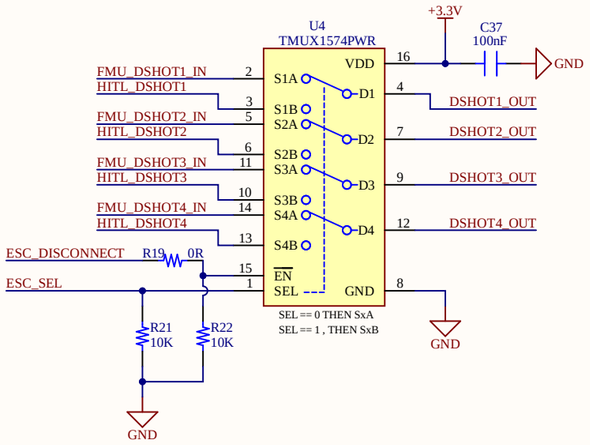
LOGIC HIGH == HITL SPOOFED SIGNAL

UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

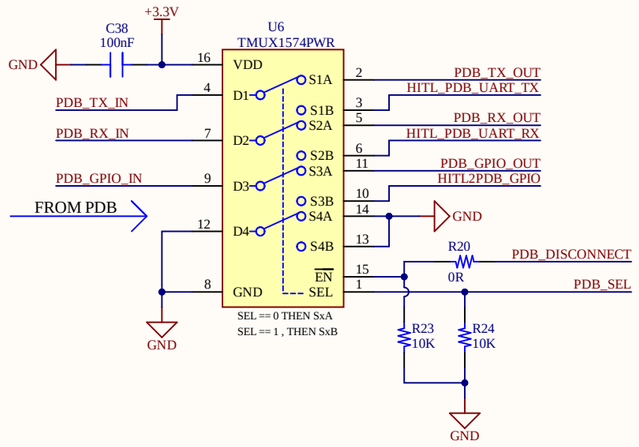
SIGNAL SPOOFING

WITH VALID MUX SUPPLY AND SEL/EN AT THEIR DEFAULT LOW
A WATCHDOG FOR MUX FAILURES WOULD BE IMPLEMENTED IN NEXT REVISION

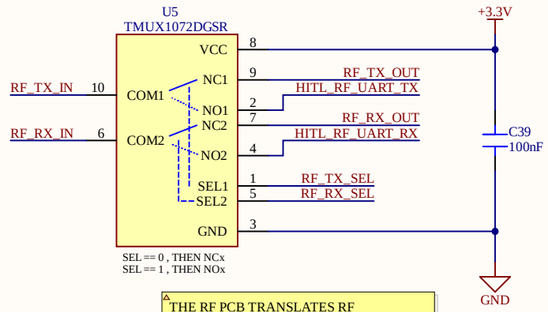
DSHOT MUX



PDB MUX



RF MUX

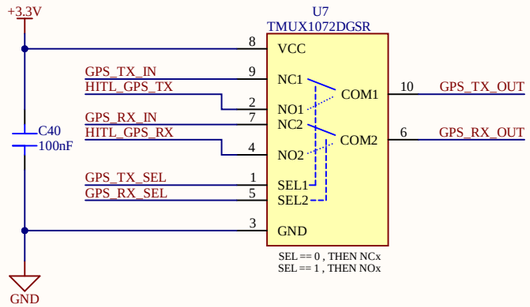


THE RF PCB TRANSLATES RF COMMUNICATION TO UART THROUGH INPUT CONNECTOR J3.

THE HITL CAN SELECT IF ITS OWN SPOOFED UART OR THE RAW ORIGINAL UART FROM RF PCB GOES THROUGH MUX.

THIS UART SIGNAL LEAVES THROUGH CONNECTOR J5 TO THE FMU

GPS MUX

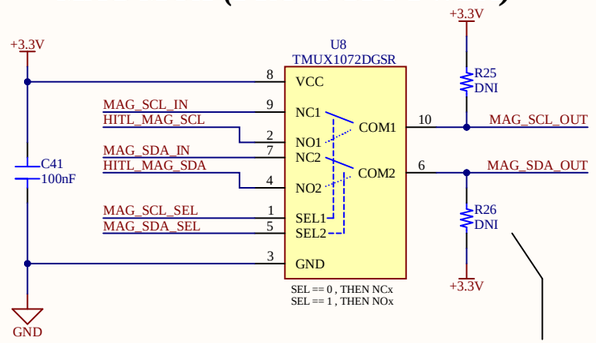


RAW GPS AND MAG DATA COMES IN THROUGH CONNECTOR P3 FROM GPS BOARD.

HITL HAS THE OPTION TO GENERATE A SPOOFED VERSION OF THIS DATA ON 2 OF ITS OWN UART PINS.

TO SEND THAT DATA AND BLOCK GPS PCB, THE HITL CONTROLS U9 TO DECIDE IF THE RAW SIGNAL OR HITL'S SPOOFED SIGNAL IS SELECTED BY U9 MUX.

MAG MUX (FROM GPS PWBA)



IF MAG I2C LINES ON GPS BOARD DO NOT HAVE PULL RESISTORS INSTALL 4.7K TO R47 AND R49

RAW GPS AND MAG DATA COMES IN THROUGH CONNECTOR P3 FROM GPS BOARD.

HITL HAS THE OPTION TO GENERATE A SPOOFED VERSION OF THIS DATA ON 2 OF ITS OWN I2C.

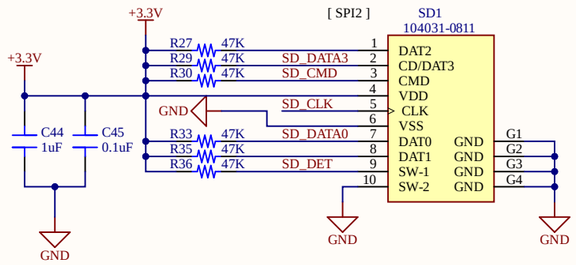
TO SEND THAT DATA AND BLOCK GPS/MAG PCB, THE HITL CONTROLS U9 TO DECIDE IF THE RAW SIGNAL OR HITL'S SPOOFED SIGNAL IS SELECTED BY U9 MUX.

SIZE	DWG NO	REV
B		-
SCALE:	VIEWDRAW	SHEET 4 OF 7

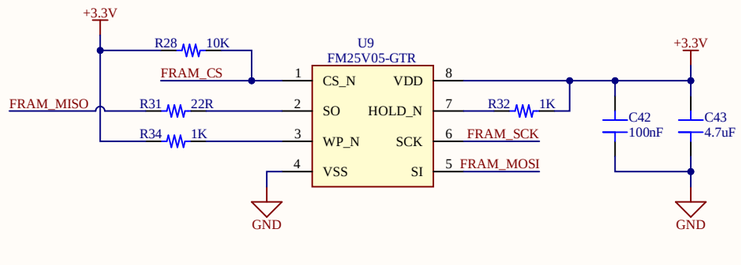
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

MEMORY

SD CARD



FRAM

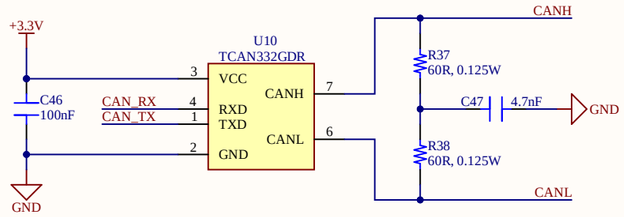


SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 5 OF 7	

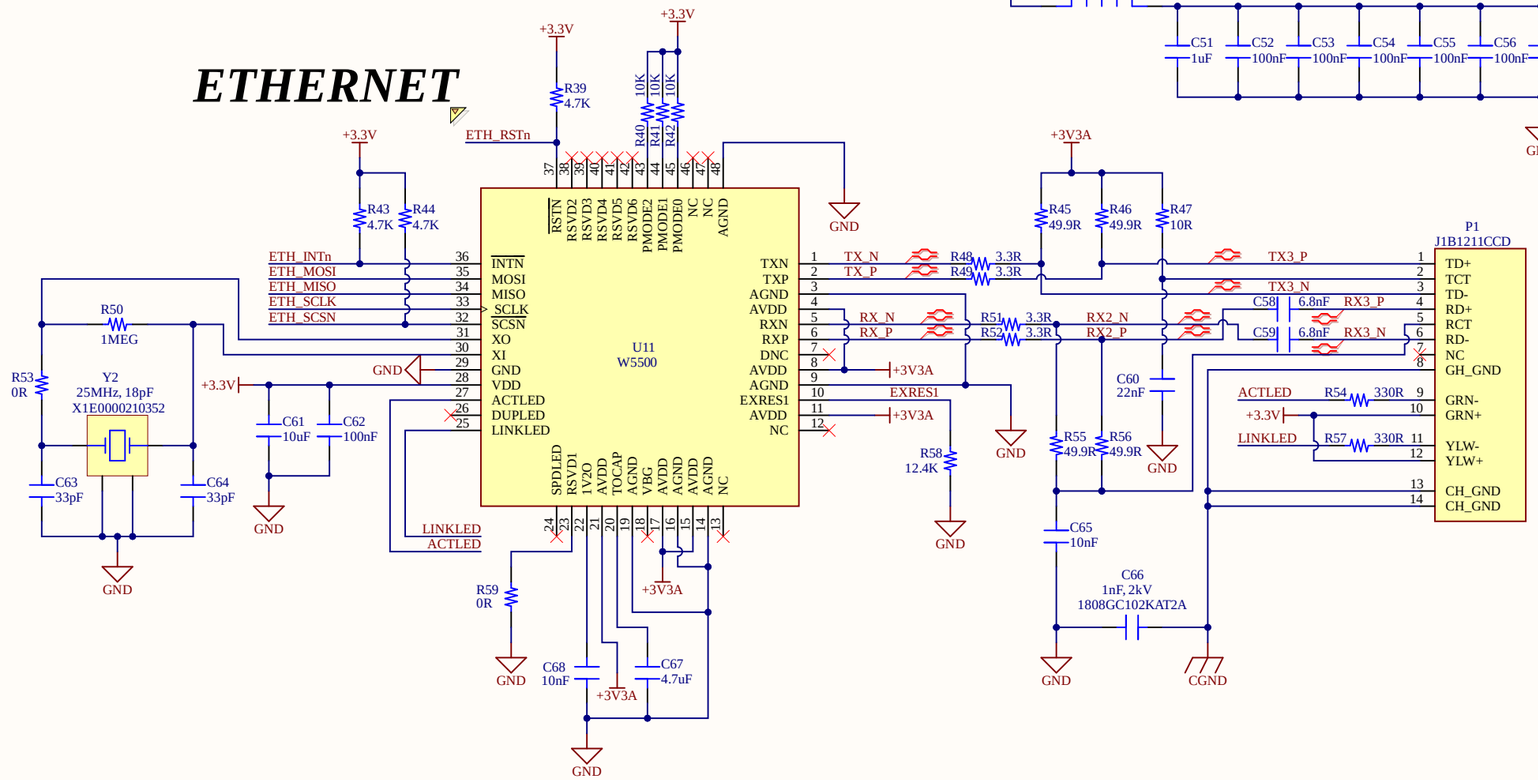
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

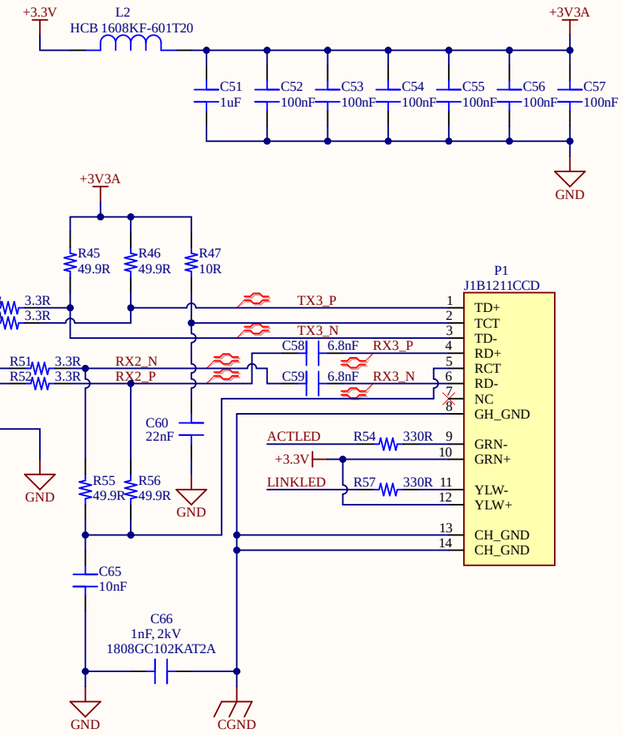
INTERFACE PT1 - CONNECTIVITY

CAN BUS



ETHERNET



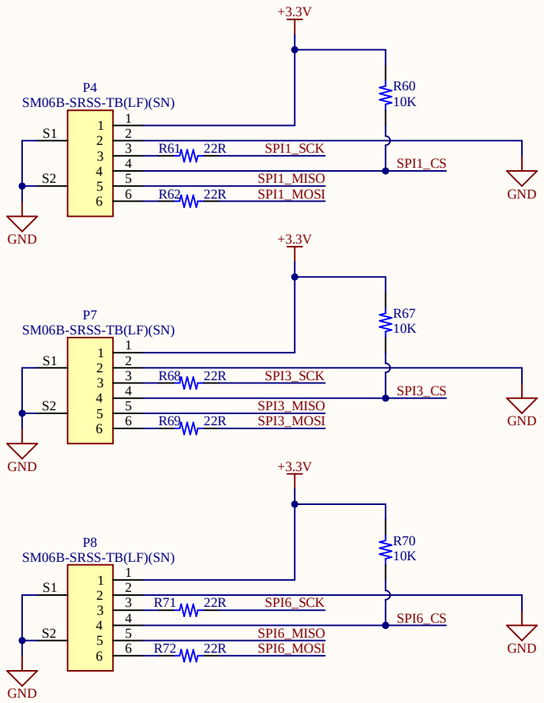


SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 6 OF 7	

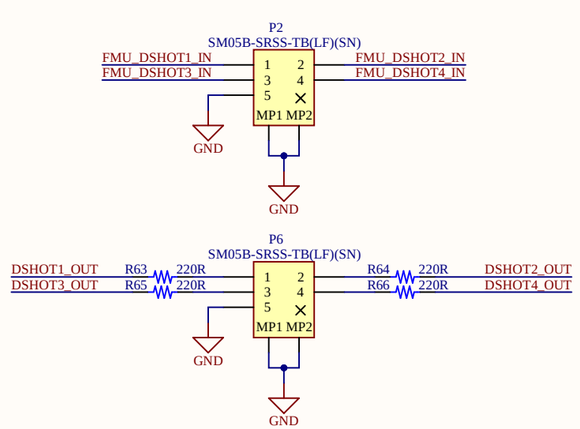
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

INTERFACE PT2 - CONNECTORS

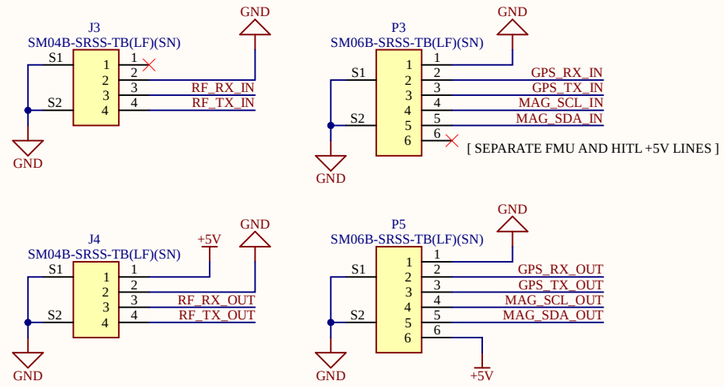
SPI CONNECTORS



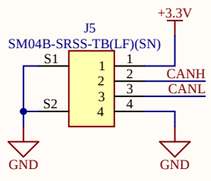
DSHOT CONNECTORS



RF CONTROLLER / GPS



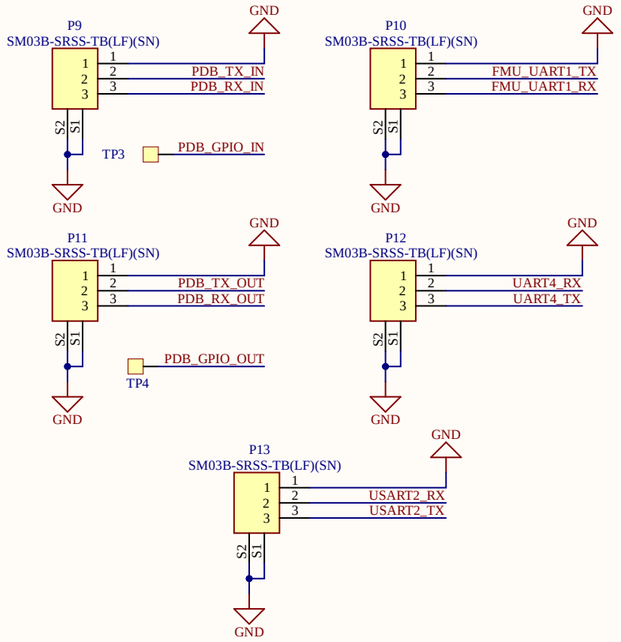
CAN BUS



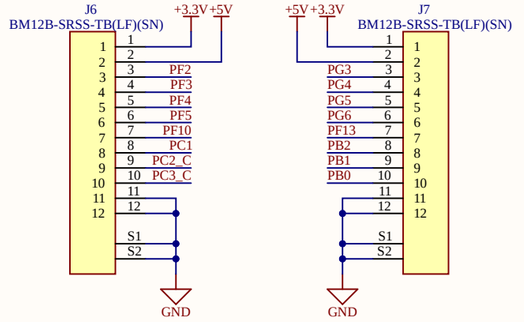
UART CONNECTORS

UART7: PDB TELEMETRY

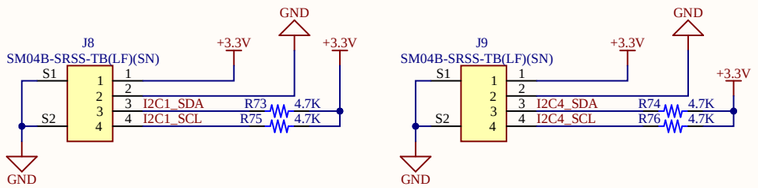
UART 1: FMU TELEMETRY



EXTRA GPIO PINS



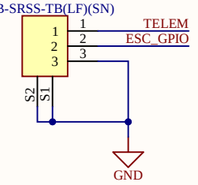
I2C CONNECTORS



CAREFUL SHARING +3.3V BETWEEN 2 +3.3V SOURCES ON 2 DIFFERENT BOARDS.

COMMS WITH ESC

THIS ENABLES HITL TO HAVE DIRECT COMMUNICATION OPTIONS WITH ESC.



EXAMPLE: If the Mux's give us some kind of issue, we can do human in the loop "hard-coded." So for the ESC, the HITL would (at the same) time pull the shared GPIO with the ESC from low to high and tell the FMU to stop DSHOT over their shared UART lines. Then the HITL can use the telem pin on the ESC to send motor instructions while also using the shared GPIO line for 2 way direct comms with ESC.

SIZE	DWG NO	REV
B		-
SCALE:	VIEWDRAW	SHEET 7 OF 7